# Budgeting & Spending — Cheat Sheet

Financial Algebra · Unit 2: Budgeting & Spending

## Essential vocabulary

**Budget**
A plan that assigns expected income to spending, saving, and debt payments for a period of time

**Cash flow**
Money coming in minus money going out during a period

**Surplus**
The amount left over when income is greater than expenses

**Deficit**
The shortfall when expenses are greater than income

**Net income (take-home pay)**
Pay remaining after taxes and other deductions; the amount a budget should be built on

**Sinking fund**
Money set aside each month for a known, irregular future expense

**Fixed expense**
A cost that stays the same amount each period

**Variable expense**
A cost whose amount changes from period to period

**Discretionary expense**
Spending on wants that could be reduced or eliminated without affecting basic needs

**Essential expense (need)**
Spending required for basic living or to keep earning income

**50/30/20 guideline**
A starting-point plan that directs about 50% of take-home pay to needs, 30% to wants, and 20% to savings and extra debt payments

**Emergency fund**
Savings reserved for unexpected expenses or income loss, often targeted at 3–6 months of essential expenses

**Financial constraint**

## Core formulas

| | |
|---|---|
| **Monthly income (biweekly)** | paycheck × $\frac{26}{12}$ |
| **Monthly income (weekly)** | paycheck × $\frac{52}{12}$ |
| **Cash flow** | income − expenses |
| **Sinking fund** | $\frac{\text{irregular bill}}{\text{months until due}}$ |
| **Percent of budget** | $\frac{\text{category}}{\text{take-home pay}} \times 100$ |
| **Annual cost of a habit** | price × times/week × 52 |
| **Break-even uses** | $\frac{\text{fixed cost}}{\text{savings per use}}$ (round up) |
| **Unit price** | $\frac{\text{price}}{\text{quantity}}$ |

## Classify every expense twice

| | |
|---|---|
| **Fixed need** | rent, car payment, insurance |
| **Fixed want** | streaming, gym contract |
| **Variable need** | groceries, utilities, gas |
| **Variable want** | dining out, shopping, events |

## 50/30/20 reference point

| | |
|---|---|
| **Needs** | ≈ 50% of take-home |
| **Wants** | ≈ 30% |
| **Savings & extra debt payments** | ≈ 20% |

A starting guideline, not a rule. High-cost areas often push needs above 50%.

A limit, such as income or a savings target, that restricts which spending choices are possible

**Savings goal**
A specific amount of money to accumulate by a target date

**Opportunity cost**
The value of the next-best alternative given up when making a choice

**Unit price**
The cost per single unit (ounce, item, load) used to compare package sizes fairly

**Impulse purchase**
An unplanned purchase made on the spot, often influenced by marketing or emotion

**Buy now, pay later (BNPL)**
A short-term financing plan that splits a purchase into installments, sometimes with late fees or interest

**Sales tax**
A percent of the purchase price collected by the state at the time of sale

## Common misconceptions

- Budgeting with gross pay instead of take-home pay.
- "Biweekly" ≠ "twice a month" (26 vs 24 checks).
- A sale is savings only if you planned to buy the item.
- A lower monthly payment does not mean a lower total cost.

## Worked example

1. Biweekly take-home \$1,300 → $1{,}300 \times \frac{26}{12}$ = \$2,816.67/month
2. Expenses \$2,640 → cash flow +\$176.67
3. Car insurance \$780 every 6 months → set aside \$130/month
4. True cash flow = 176.67 − 130 = +\$46.67 (thin margin!)

## Questions to ask before deciding

- Is this a need or a want — and who decides?
- What will this cost per year?
- What am I giving up (opportunity cost)?
- Which irregular bills are coming in the next 12 months?
- What will I do if income drops for a month?

## Decision tips

- Pay yourself first: automate savings on payday.
- Round hours and months UP; round maximum spending DOWN.
- Review subscriptions every 3 months.
- Keep a starter emergency fund before extra spending.

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